



Technische Hochschule Deggendorf

Case study project

Case study Assistive Robotics for improvement of Life Quality

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Abstract

Telepresence robots play an important role in assisting elderly people by enabling remote communication and safe indoor navigation. This project presents a LiDAR-Based Telepresence Robot for Elderly Assistance Using Voice-Controlled Autonomous Navigation. The system allows a family member, such as a grandson, to remotely control the robot using voice commands. The robot uses LiDAR-based obstacle detection and path planning to navigate safely to the grandmother's location.

The system is developed using ROS 2, Flask, Python and SQLite. It provides an interactive web interface for voice-based navigation and real-time robot monitoring. By combining telepresence, autonomous navigation, and voice control, the project offers an effective assistive solution for elderly care. The proposed system can handle with Manually and voice assistance

1. Introduction

Telepresence robots are becoming an important technology in healthcare and elderly assistance, enabling people to communicate and interact remotely without being physically present. They are especially useful for elderly individuals who live alone or have limited mobility, as they allow family members and caregivers to stay connected and monitor their well-being from any location. By combining communication with autonomous navigation, telepresence robots improve accessibility, provide timely assistance, and help reduce social isolation among elderly people.

This project presents a **LiDAR-Based Telepresence Robot for Elderly Assistance Using Voice-Controlled Autonomous Navigation**. The system allows a family member, such as a grandson, to remotely control the robot using voice commands through a web interface. The robot uses LiDAR technology to detect obstacles and navigate safely to the grandmother's location. The project is developed using Python, Flask, HTML, CSS, JavaScript, and SQLite, providing an interactive and user-friendly platform for robot control and monitoring. The proposed system demonstrates how telepresence and autonomous navigation can be integrated to support elderly care and can be further extended to real-world robotic applications.

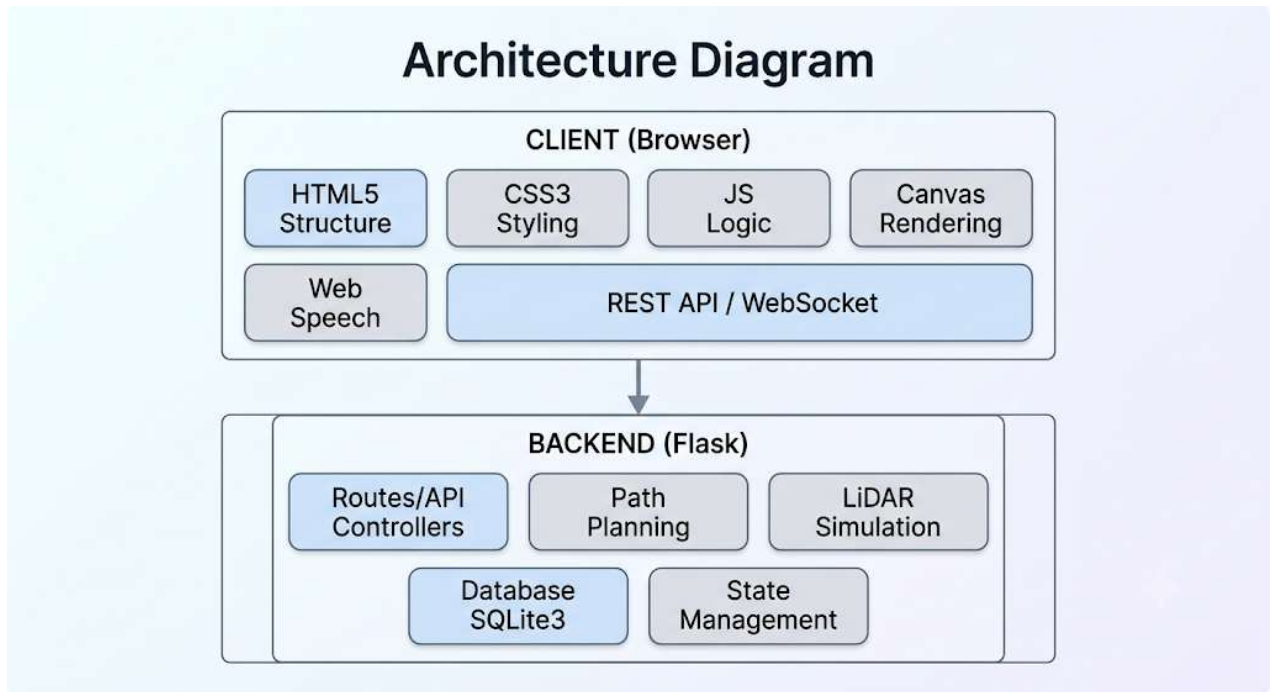
1.1 Background Work

Autonomous robot navigation has emerged as one of the most critical areas in robotics research, with applications ranging from industrial automation to domestic assistance. The integration of Simultaneous Localization and Mapping (SLAM), path planning algorithms, and sensor fusion techniques has revolutionized how robots perceive and interact with their environment. LiDAR (Light Detection and Ranging) technology has become particularly significant in this domain, offering precise distance measurements and obstacle detection capabilities essential for safe autonomous navigation.

The Robot Operating System (ROS) has established itself as the de facto standard framework for robotic applications, providing a flexible architecture for developing complex robotic systems. ROS enables communication between various components, including sensors, actuators, and planning algorithms, through a publish-subscribe messaging system. However, the complexity and resource requirements of full ROS implementations often limit accessibility for educational purposes and rapid prototyping.

Web-based simulation environments have gained prominence as they offer platform-independent, accessible, and visually intuitive interfaces for understanding complex robotics concepts. These simulations bridge the gap between theoretical knowledge and practical implementation, allowing students, researchers, and enthusiasts to experiment with navigation algorithms without the need for expensive hardware.

1.2 Architecture Diagram



2. Problem Statement and Measurable Outcomes

Conventional telepresence robots are typically controlled using a mobile phone, laptop, or joystick, requiring users to manually operate the robot through buttons or touch controls. This method can be inconvenient, especially when quick and hands-free operation is needed. To overcome this limitation, the proposed system introduces a voice-controlled telepresence robot that allows the user, such as a grandson, to control the robot using simple voice commands like "Go to the kitchen" or "Go to Grandma's room." The robot processes the voice command, determines the destination, and autonomously navigates to the specified location using LiDAR-based obstacle detection and path-planning techniques while avoiding obstacles. The success of the system is measured by its ability to accurately recognize voice commands, safely navigate to the desired location, avoid collisions, and provide a simple, efficient, and user-friendly method of remote integration for elderly assistance.

2.1 Existing Systems

System	Features	Platform
Gazebo	Full physics simulation, ROS integration	Desktop/Linux
Webots	Professional robotics simulator	Cross-platform
V-REP/CoppeliaSim	Modular robotics simulation	Cross-platform
TurtleBot Simulator	ROS-based mobile robot simulation	ROS/Ubuntu
MATLAB Robotics Toolbox	Algorithm development and simulation	MATLAB

Table 1: system, Features and platforms used

2.2.Drawbacks

1. **Resource Intensive:** Systems like Gazebo demand substantial computational resources, limiting accessibility on standard hardware.
2. **Limited Web Accessibility:** Most existing simulators are desktop applications, restricting access to specific devices.
3. **Steep Learning Curve:** Understanding ROS concepts, message passing, and parameter configuration requires extensive documentation study.

4. **Lack of Voice Interaction:** Many existing systems do not incorporate natural language processing for robot command input.
5. **Visualization Limitations:** Traditional simulators often lack modern, intuitive user interfaces with real-time feedback.
6. **Cost Factors:** Commercial simulators like Webots and CoppeliaSim require licenses for full functionality

3. Project Objectives

The primary objectives of this project are summarized below.

Objective ID	Description
O1	Develop a voice-controlled telepresence robot for remote elderly assistance.
O2	Enable the robot to recognize voice commands
O3	Enable the robot to recognize voice commands
O4	User Friendly assistive Robotic System

Table 2: Project Objectives and Description

4. Literature Review Summary

The literature review analyzed recent research on deep learning-based surface defect detection. Studies highlight the limitations of anchor-based detection models and emphasize the need for lightweight, real-time solutions.

Criteria	Research Insight	Proposed Model Justification
Navigation	SLAM Required	SLAM based Mapping
Localization	AMCL Used	Accurate Positioning
Path Planning	A* algorithm	Efficient navigation
Safty	Obstacle avoidance	Real – time safty

Table 3: Alignment of reviewed research insights with the proposed model selection.

5. System Development and Environment Support

The telepresence robot system was developed to provide a simple and efficient method for remotely controlling the robot using voice commands. The system integrates a Flask-based backend with a web interface developed using HTML, CSS, and JavaScript. SQLite is used to store robot status and navigation information, while Python manages voice command processing and robot control. LiDAR-based navigation enables the robot to detect obstacles

5.1 Development Environment

Component	Technology Used
Programming Language	Python
Python	Flask
Frontend	ROS2
Database	SQLite

Navigation	LiDAR
Voice Control	Speech Recognition

Table 4: Development Environment and Technologies Used

5.2 System Design

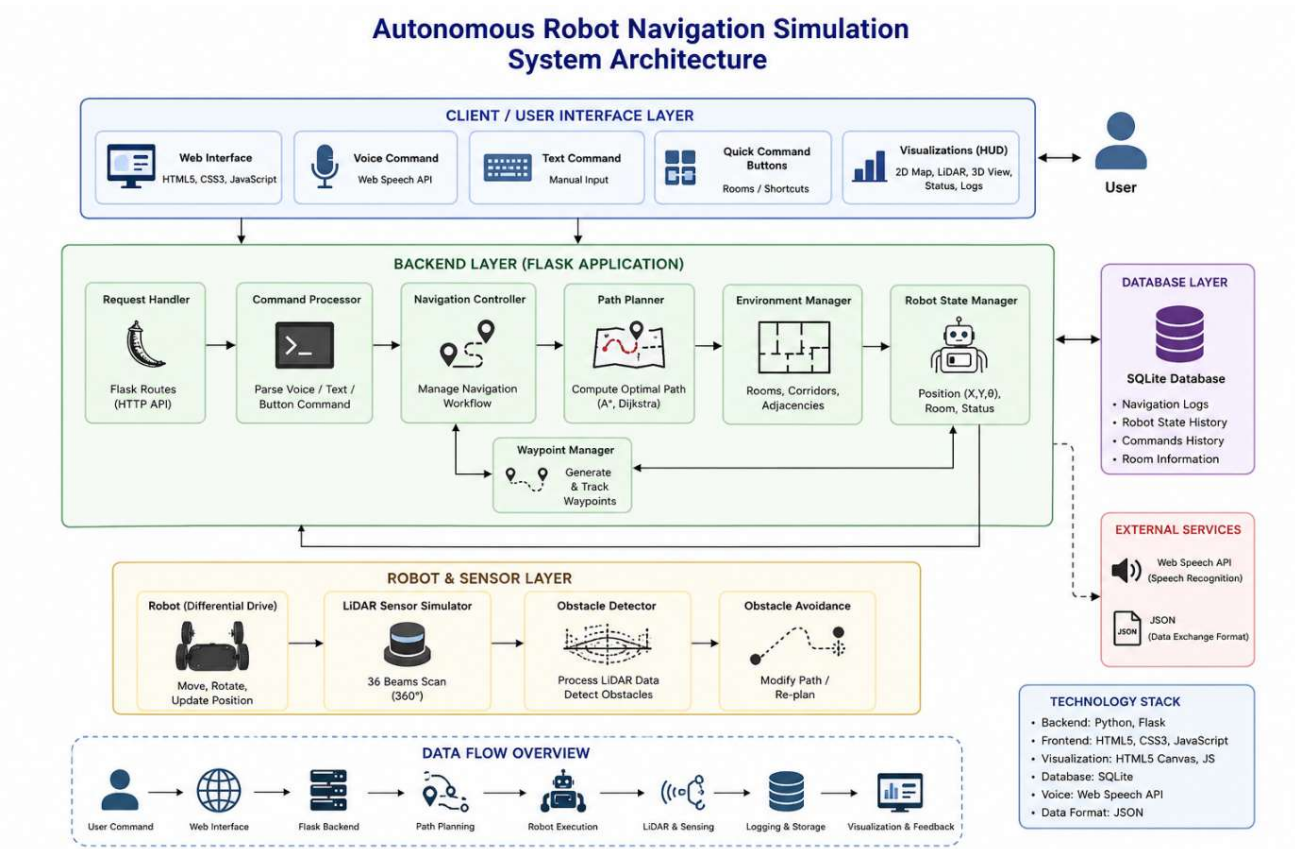


Figure 1: Autonomous Robot System Architecture

6. Voice Command Processing and Navigation Configuration

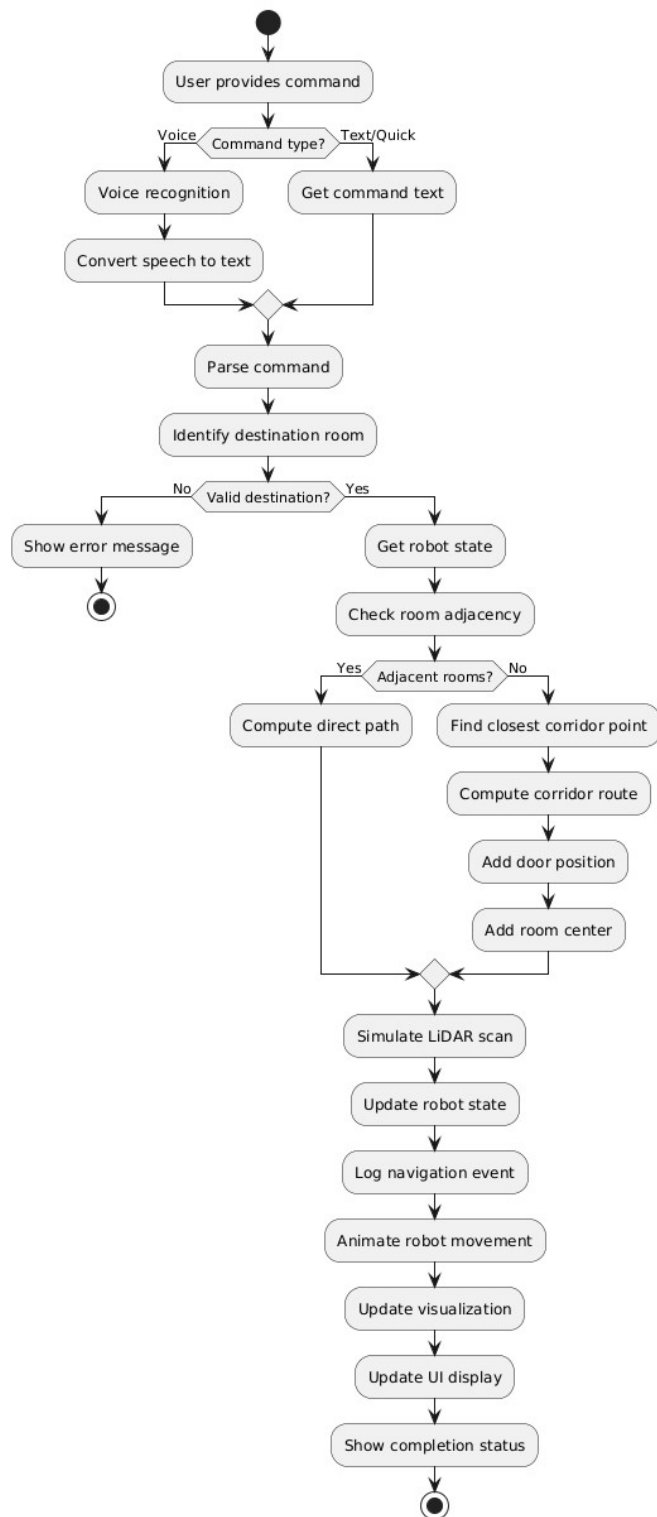
The system processes voice commands given by the user to control the telepresence robot. Commands such as "Go to the kitchen" or "Go to Grandma's room" are converted into text using a speech recognition module. The backend interprets the command, identifies the destination, and generates a navigation request for the robot.

The robot uses LiDAR-based navigation to detect obstacles and determine a safe path to the selected destination. During movement, the robot continuously updates its route to avoid obstacles and safely reach the target location. This approach provides hands-free robot control and improves the overall user experience for remote elderly assistance.



Figure 2: Voice Command Processing and Robot Navigation Workflow

7. Model Training



8. Future Work

Short-term enhancements for the ROS LiDAR Robot Navigation System include expanding the platform to support multiple robots with collision avoidance, introducing dynamic obstacles with real-time path recalculation, integrating advanced SLAM for mapping unknown environments, and incorporating machine learning techniques such as reinforcement learning to enable adaptive navigation behavior. In the long term, the system can be further evolved by connecting it to real hardware robots through ROS Bridge or MQTT for physical deployment, extending the environment into full 3D navigation spaces, and developing a companion mobile application for remote control and monitoring. Additional possibilities include integrating real-time camera feeds for visual odometry and object detection, enabling multi-floor navigation across complex building structures using elevators or staircases, implementing fleet management algorithms for coordinated multi-robot systems, and leveraging cloud-based infrastructure to support scalable, distributed, and remotely accessible robotic operations.

9. Applications

The telepresence robot with integrated navigation system has a wide range of applications across education, healthcare, industry, and research domains. In education, it enables remote learning experiences where students or instructors can attend classrooms and labs virtually while controlling the robot's movement in real time. In healthcare, it can be used for remote doctor consultations, patient monitoring, and hospital rounds, reducing physical presence while maintaining interaction quality. In industrial and corporate environments, the system supports remote site inspection, office presence, and collaboration, allowing employees to navigate and communicate from distant locations. It is also valuable for security and surveillance applications, enabling remote patrols of restricted or hazardous areas. From a research perspective, it can be used to study human-robot interaction, autonomous navigation in shared spaces, and teleoperation systems combining manual and assisted control. Overall, the system enhances accessibility, safety, and efficiency by enabling users to physically "be present" in remote environments through real-time navigation and communication.

10 . Limitations

The telepresence robot with an integrated navigation system also has several limitations that should be acknowledged. One major limitation is its dependency on a stable internet connection, as any latency, jitter, or disconnection can severely affect real-time control, video streaming, and navigation responsiveness. In complex or highly dynamic environments, the navigation system may struggle with accurate obstacle detection and path planning, especially if sensor data (like LiDAR or camera input) is noisy or limited. The system may also face challenges in fully autonomous decision-making, often requiring human intervention in unfamiliar or cluttered spaces. Hardware constraints such as limited battery life, processing power, and sensor range can further restrict operational time and accuracy. Additionally, telepresence robots may suffer from reduced situational awareness compared to physical presence, as users rely only on camera and sensor feeds, which can miss peripheral or occluded information. Cost and maintenance of hardware components, along with calibration and setup complexity, can also be barriers for large-scale deployment. Finally, privacy and security concerns may arise, particularly in sensitive environments where remote access and live video transmission are involved.

11. Conclusion

The ROS LiDAR Robot Navigation System successfully demonstrates an interactive web-based autonomous navigation simulation that integrates essential robotics principles including path planning, LiDAR-based obstacle detection, and human-robot interaction in a cohesive and visually engaging environment. It achieves efficient room-to-room navigation through intelligent adjacency-based path planning with corridor routing, while a real-time 36-beam LiDAR simulation provides accurate obstacle awareness represented in both polar and Cartesian formats. The system further enhances usability through a multi-modal interface that supports voice commands, text input, and quick navigation controls, making robot interaction intuitive and flexible. A cyberpunk-inspired HUD with synchronized 2D and 3D visualizations improves interpretability of navigation behavior, while SQLite-based persistence ensures continuity of robot state, sensor readings, and navigation history across sessions. Overall, the project not only meets its functional objectives but also serves as an effective educational platform for demonstrating ROS concepts such as topics, messages, and navigation stacks within a scalable, high-performance web architecture capable of maintaining smooth real-time updates and broad cross-platform compatibility.

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